Section A — Concept Application

1) Question: What factors should be considered when selecting the CPU, RAM, and storage to ensure good performance and hardware compatibility?

For a typical office computer used for browsing, Microsoft Office, IT support, and multitasking:

- CPU: Modern Intel Core i5 or AMD Ryzen 5.
- RAM: 16 GB DDR4 or DDR5, depending on motherboard support.
- Storage: 512 GB NVMe SSD, if supported.
- Motherboard: Must support the selected processor, RAM type, and storage interface

2) Explain how the antivirus software affects CPU, RAM, and storage performance, leading to system slowdowns.

1. Impact on CPU, RAM, and storage

Component	How antivirus affects it	Possible slowdown
CPU	Uses processing power to scan files, detect malware, and analyze suspicious programs.	Applications may open slowly, and the computer may become less responsive during scans.
RAM	Uses memory to run background services, maintain security databases, and monitor applications.	Less memory remains available for other programs, potentially causing lag or increased paging.
Storage (HDD/SSD)	Reads files during scans, writes logs and updates, and checks files when they are opened or downloaded.	File copying, application loading, and disk operations may become slower, especially on a busy or slow HDD.

2. Example scenario

Suppose you are working on a computer with 8 GB RAM and an HDD.

- You start a full antivirus scan.
- CPU usage increases as the antivirus examines files.
- RAM usage rises because antivirus services and other applications are running simultaneously.
- Disk activity increases as thousands of files are read and checked.
- Opening applications or copying files may take longer than usual.

The slowdown is often temporary and may reduce after the scan finishes.

3. How to reduce antivirus-related slowdowns

1. Schedule full scans during breaks or outside working hours.
2. Keep antivirus software updated to benefit from performance improvements.

3. Avoid running multiple real-time antivirus products simultaneously.
4. Check Task Manager (Ctrl + Shift + Esc) to identify whether antivirus processes are using excessive CPU, memory, or disk resources.
5. Upgrade hardware if needed: adding RAM or replacing an HDD with an SSD can improve overall responsiveness.

3) Describe the steps you would follow to troubleshoot the problem if it is caused by driver incompatibility or operating system issues.

1. Identify the Problem

- Ask the user when the problem started and whether any new software, drivers, or Windows updates were installed.
- Note symptoms such as slow performance, system crashes, error messages, or devices not working correctly.

2. Check Device Manager

- Press Windows + X and select Device Manager.
- Look for devices with a yellow warning symbol.
- Identify outdated, missing, or incompatible drivers.

3. Update or Roll Back Drivers

- Right-click the affected device and select Update driver.
- Download compatible drivers from the computer or hardware manufacturer's official website if needed.
- If the problem started after a driver update, open Properties → Driver → Roll Back Driver, if available.

4. Check Windows Updates

- Open Settings → Windows Update.
- Check for pending updates and install appropriate updates.
- Restart the computer and check whether the issue is resolved.

5. Scan and Repair System Files

- Open Command Prompt or Terminal as an administrator.
- Run the following commands one at a time

4) Explain how a UPS and surge protector help protect computer hardware and prevent data loss during these situations

1. UPS (Uninterruptible Power Supply)

A UPS provides temporary battery power when the main electricity supply fails.

- Prevents sudden shutdowns: Gives the user time to save files and shut down the computer safely.
- Reduces data loss: Helps prevent unsaved work from being lost when the power goes out.
- Protects hardware: Many UPS models provide voltage regulation to help protect computers from voltage fluctuations.
- Supports continuity: Keeps the computer running briefly during a power cut, depending on battery capacity and connected load.

2. Surge Protector

A surge protector helps protect computer equipment from sudden increases in voltage.

- Protects components: Helps prevent electrical surges from damaging the motherboard, power supply, monitor, and other connected devices.
- Reduces electrical damage: Limits or diverts excess voltage, depending on its design and protection rating.
- Protects connected equipment: Can protect devices such as routers, printers, and monitors when connected to appropriately rated protection.

5) DNS (Domain Name System) and DHCP (Dynamic Host Configuration Protocol) work together to help devices connect to network resources using domain names instead of IP addresses.

1. How DNS and DHCP Work Together

- DHCP: Automatically assigns a device an IP address, subnet mask, default gateway, and DNS server address when it joins the network.
- DNS: Translates domain names, such as www.google.com, into IP addresses so the device can connect to the correct server.

Example: When a computer connects to office Wi-Fi, DHCP assigns its network settings, including the DNS server address. When the user opens a website, DNS resolves the website's name to an IP address, allowing the browser to connect.

6) Describe the troubleshooting steps you would take to determine whether the firewall rules are blocking legitimate traffic, and how you would resolve the issue.

1. Identify the Problem

- Ask the user which application, website, or network service is not working.
- Record the affected computer's IP address, destination IP address, port number, and protocol (TCP or UDP).
- Check whether the problem affects one computer or multiple devices.

2. Test Network Connectivity

Run these commands in Windows Command Prompt:

ipconfig /all

ping 192.168.1.1

nslookup www.google.com

- ipconfig /all checks the computer's IP configuration and DNS settings.
- ping checks connectivity to the default gateway.
- nslookup checks whether DNS can resolve a domain name.

If DNS and basic connectivity work but a particular application still fails, a firewall rule may be blocking its traffic.

3. Check Firewall Rules

- Open Windows Security → Firewall & network protection → Advanced settings.
- Review Inbound Rules and Outbound Rules.
- Look for rules that block the affected application, port, protocol, or destination.
- Check whether the rule applies to the active network profile: Domain, Private, or Public.

4. Check Firewall Logs

Review firewall logs for blocked connections, including the source and destination IP addresses, port numbers, and timestamps.

For Windows Defender Firewall, logging can be configured through the firewall's advanced settings. On a business firewall, review its traffic or security logs.

5. Test the Required Port

If the application uses a known TCP port, test it in PowerShell:

SECTION B — PRACTICAL APPLICATION

Section B — Practical Coding Tasks







Section D — AI-Augmented Learning

1) The AI-generated plan must specify the subnet, subnet mask, usable host range, and broadcast address.

STEP 1 · BUILD WITH AI

Parameter Value

Network subnet 192.168.1.0/24 Subnet mask 255.255.255.0

Usable host range 192.168.1.1 – 192.168.1.254

Broadcast address 192.168.1.255

Total IP addresses 256

Usable host addresses 254

Explanation:

- Subnet: Identifies the network segment.

- Subnet mask: Determines which bits represent the network and which represent hosts.

- Usable host range: The IP addresses that can be assigned to devices, such as computers, printers, and servers.

- Broadcast address: Used to send a packet to all devices on the subnet.

Example allocation:

- Router/default gateway: 192.168.1.1

- Computers: 192.168.1.10 – 192.168.1.100

- Printers and servers: 192.168.1.101 – 192.168.1.120

- DHCP pool: 192.168.1.121 – 192.168.1.254

This plan supports up to 254 usable IPv4 addresses on the office subnet

2) The plan must allocate addresses for all 5 systems and 1 printer named in the scenario.

Address Allocation Plan for 5 Systems and 1 Printer

Using the subnet 192.168.1.0/24, the following plan assigns a unique static IP address to each of the five systems and the printer.

Device	IP address	Subnet mask	Default gateway
Router	192.168.1.1	255.255.255.0	—
System 1	192.168.1.10	255.255.255.0	192.168.1.1
System 2	192.168.1.11	255.255.255.0	192.168.1.1
System 3	192.168.1.12	255.255.255.0	192.168.1.1
System 4	192.168.1.13	255.255.255.0	192.168.1.1
System 5	192.168.1.14	255.255.255.0	192.168.1.1
Printer	192.168.1.20	255.255.255.0	192.168.1.1

3) Instruct the AI to also identify which addresses to reserve for future expansion.

Device / Purpose	IP address or range
Network address	192.168.1.0
Router / Default gateway	192.168.1.1
System 1	192.168.1.10
System 2	192.168.1.11
System 3	192.168.1.12
System 4	192.168.1.13
System 5	192.168.1.14
Printer	192.168.1.20
Future expansion — reserved	192.168.1.21 – 192.168.1.49
Other devices / DHCP pool	192.168.1.50 – 192.168.1.254
Broadcast address	192.168.1.255

STEP 2 · TEST & DEBUG

Overlapping ranges: No overlap exists between the assigned static addresses, the reserved range, and the proposed DHCP pool.

1) Network and broadcast addresses: 192.168.1.0 is the network address and 192.168.1.255 is the broadcast address. Neither is assigned to a device.

2) Host capacity: A /24 subnet supports 254 usable host addresses, which is sufficient for the current six devices and future growth.

3) Future expansion: The range 192.168.1.21–192.168.1.49 reserves 29 addresses for future devices.

4) DHCP configuration: The DHCP pool must exclude all static and reserved addresses. If the DHCP pool extends to .254, the reserved .21–.49 range remains excluded.